

FloodEye

AWS Production Upgrade Costing Tables

Dual-Currency Cost Analysis: USD (\$) & INR (INR @ 84.50/\$)

1. Itemized Monthly Upgrade Budget (50-Node Production Pilot)

Upgrade Pillar	AWS Service	Specification / Sizing	Unit (\$)	USD/mo	INR/mo
MQTT Protocol	AWS IoT Core	8.64M msgs/month	\$1.00 / M	\$10.00	845
PostgreSQL DB	RDS db.t4g.small	2 vCPU, 2 GB, 50 GB SSD	On-Demand	\$34.00	2,873
Multi-Node App	ECS Fargate + ALB	2 tasks, 0.5 vCPU each	Per-vCPU/hr	\$28.00	2,366
SMS Gateway	SNS + Lambda + Twilio	5k SMS + 100K invokes	\$0.005 / SMS	\$25.00	2,113
ML Inference	SageMaker Serverless	100K reqs, 2 GB mem	Per-invoke	\$15.00	1,268
CDN / Network	CloudFront + Route53	100 GB out, 1M WAF req	\$0.085 / GB	\$12.00	1,014
Cache Layer	ElastiCache Redis	t4g.micro (0.5 GB RAM)	On-Demand	\$12.00	1,014
Monitoring	CloudWatch + X-Ray	10 GB logs/month	\$0.50 / GB	\$8.00	676
Total Monthly Upgrade Cost – 50-Node Production Pilot				\$144.00	12,168
Annualized Upgrade Budget (12 Months)				\$1,728.00	1,46,016

Table 1: Itemized Monthly AWS Upgrade Infrastructure Budget (ap-south-1 Mumbai, July 2025)

2. Multi-Tier Scale-Up Cost Projections (50 to 2,000 Nodes)

Deployment Tier	Nodes	Msgs/mo	Database Sizing	USD/mo	INR/mo
Pilot (River Basin)	50	8.64M	db.t4g.small (2 vCPU, 2 GB)	\$144	12,168
District (Kamrup)	200	34.56M	db.t4g.medium (2 vCPU, 4 GB)	\$380	32,110
State-Wide (Assam)	500	86.40M	db.r7g.large (2 vCPU, 16 GB)	\$920	77,740
National Grid (India)	2,000	345.60M	db.r7g.xlarge + Read Replicas	\$3,200	2,70,400

Table 2: Upgrade Cost Projections across Network Deployment Tiers (USD & INR)

3. Net Upgrade Cost Comparison (Unoptimized vs. Optimized)

Budget Scenario	Optimization Basis	Monthly (USD)	Monthly (INR)
Unoptimized Baseline	Standard On-Demand AWS pricing (50 Nodes)	\$144.00	INR 12,168
Optimized Pilot Cost	With Reserved Instances, Glacier, CBOR & Seasonal ML	\$95.00	INR 8,028
Net Monthly Savings	Total recurring cost reduction (34% savings)	-\$49.00	-INR 4,140

Table 3: Net Upgrade Cost Comparison: Unoptimized vs. Optimized Budget (USD & INR)